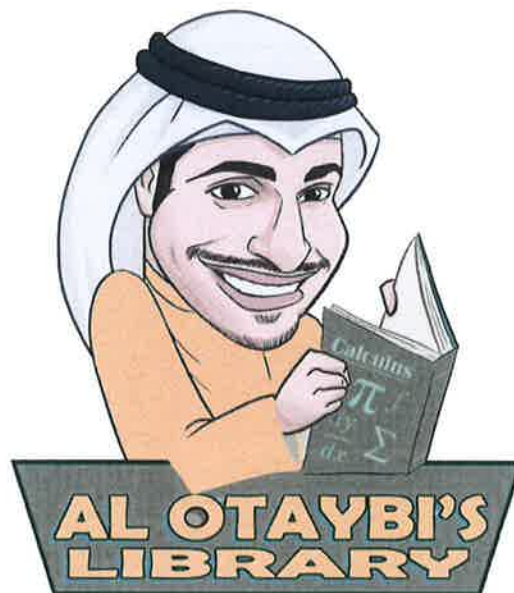


Kuwait University

2021/2022

Spring course



Calculus B

Second

Section : 11.10

11.10 Taylor and Maclaurin Series.

(14)



11.10 Power series representation



Form	Representation	Radius of convergence
e^u	$\sum_{n=0}^{\infty} \frac{u^n}{n!}$	∞
$\sin(u)$	$\sum_{n=0}^{\infty} (-1)^n \frac{u^{2n+1}}{(2n+1)!}$	∞
$\cos(u)$	$\sum_{n=0}^{\infty} (-1)^n \frac{u^{2n}}{(2n)!}$	∞
$\sinh(u)$	$\sum_{n=0}^{\infty} \frac{u^{2n+1}}{(2n+1)!}$	∞
$\cosh(u)$	$\sum_{n=0}^{\infty} \frac{u^{2n}}{(2n)!}$	∞



Do NOT forget



قوانين الدوال المثلثية

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$\sin 2x = 2 \sin x \cos x$$

$$\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$$

$$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$$



قوانين الهايبربوليك

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\sinh 2x = 2 \sinh x \cosh x$$



قوانين اللوق و a^x

$$\log_a x = \frac{1}{\ln a} \ln x$$

$$a^x = e^{\ln a^x} = e^{x \ln a}$$

Find the power series representation of

$$f(x) = \int \frac{x}{e^x} dx$$

$$f(x) = \int x e^{-x} dx$$

$$= \int x \sum_{n=0}^{\infty} \frac{(-x)^n}{n!} dx$$

$$= \int x \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!} dx$$

$$= \int \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n!} dx$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+2}}{(n+2) n!} + C$$

$$\therefore \int x e^{-x} dx = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+2}}{(n+2) n!} + C$$

$$\underline{\underline{x=0}}$$

$$0 = 0 + C \rightarrow C = 0$$

$$\therefore \int x e^{-x} dx = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+2}}{(n+2) n!}$$

radius of convergence:
 $R = \infty$

Find the power series representation of

$$f(x) = \int_0^x \frac{\sinh t}{t} dt$$

$$f(x) = \int_0^x \frac{1}{t} \sinh t \, dt$$

$$= \int_0^x \frac{1}{t} \sum_{n=0}^{\infty} \frac{t^{2n+1}}{(2n+1)!} dt$$

$$= \int_0^x \sum_{n=0}^{\infty} \frac{t^{2n}}{(2n+1)!} dt$$

$$= \sum_{n=0}^{\infty} \frac{[t^{2n+1}]_0^x}{(2n+1)(2n+1)!}$$

$$\Rightarrow f(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)(2n+1)!}$$

$$\Rightarrow R = \infty$$

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Find the Taylor series for

$$f(x) = \frac{1}{x} \quad \text{at } a = 3$$

$$x - 3 = 0$$

$$\text{Let } t = x - 3 \longrightarrow x = 3 + t$$

$$\begin{aligned} \Rightarrow f(t) &= \frac{1}{3+t} \\ &= \frac{1}{3} \cdot \frac{1}{1 + \frac{t}{3}} \\ &= \frac{1}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{t}{3}\right)^n \\ &= \frac{1}{3} \sum_{n=0}^{\infty} (-1)^n \frac{t^n}{3^n} \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{t^n}{3^{n+1}} \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{(x-3)^n}{3^{n+1}} \end{aligned}$$

$$\begin{aligned} \Rightarrow \left| \frac{t}{3} \right| &< 1 \\ |t| &< 3 \end{aligned}$$

$$|x-3| < 3 \longrightarrow -3 < x-3 < 3 \longrightarrow 0 < x < 6 \longrightarrow R = \frac{6}{2} = 3$$

Find the power series representation for

$$f(x) = \frac{2}{3x - 4} \text{ about } x = 2$$

and determine the radius of convergence

$$x - 2 = 0$$

$$\text{Let } t = x - 2 \longrightarrow x = t + 2$$

$$f(t) = \frac{2}{3(t+2) - 4} = \frac{2}{3t + 6 - 4}$$

$$= \frac{2}{3t + 2} = \frac{2}{2 \left[1 + \frac{3t}{2} \right]}$$

$$= \frac{1}{1 + \frac{3t}{2}} = \sum_{n=0}^{\infty} (-1)^n \left(\frac{3t}{2} \right)^n$$

$$= \sum_{n=0}^{\infty} (-1)^n \left(\frac{3(x-2)}{2} \right)^n$$

$$\Rightarrow \left| \frac{3}{2}t \right| < 1 \longrightarrow |t| < \frac{2}{3} \longrightarrow |x-2| < \frac{2}{3}$$

$$R = \frac{2}{3}$$

Find the power series representation of

$$f(x) = \ln(1+x) \text{ about } x=1$$

$$x-1=0$$

$$t = x-1 \longrightarrow x = t+1$$

$$f(t) = \ln(1+t+1) = \ln(2+t)$$

$$= \ln \left[2 \left(1 + \frac{t}{2} \right) \right]$$

$$= \ln 2 + \ln \left(1 + \frac{t}{2} \right)$$

$$= \ln 2 + \sum_{n=0}^{\infty} \frac{(-1)^n \left(\frac{t}{2} \right)^{n+1}}{n+1}$$

$$= \ln 2 + \sum_{n=0}^{\infty} \frac{(-1)^n \left(\frac{x-1}{2} \right)^{n+1}}{n+1}$$



$$\left| \frac{t}{2} \right| < 1$$

$$\left| \frac{x-1}{2} \right| < 1$$

$$|x-1| < 2$$

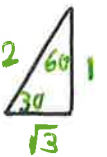
$$R = 2$$

Find the Taylor series of $f(x) = \sin x$ at $x = \frac{\pi}{6}$

$$x - \frac{\pi}{6} = 0$$

$$\text{Let } t = x - \frac{\pi}{6} \longrightarrow x = t + \frac{\pi}{6}$$

$$f(t) = \sin\left(t + \frac{\pi}{6}\right)$$



$$= \sin(t) \cos \frac{\pi}{6} + \cos(t) \sin \frac{\pi}{6}$$

$$= \sin(t) \frac{\sqrt{3}}{2} + \cos(t) \frac{1}{2}$$

$$= \frac{\sqrt{3}}{2} \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n+1}}{(2n+1)!} + \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n}}{(2n)!}$$

$$= \frac{\sqrt{3}}{2} \sum_{n=0}^{\infty} (-1)^n \frac{\left(x - \frac{\pi}{6}\right)^{2n+1}}{(2n+1)!} + \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \frac{\left(x - \frac{\pi}{6}\right)^{2n}}{(2n)!}$$

$$\Rightarrow R = \infty$$



Text book questions



Find the Maclaurin series of the given function

$$f(x) = x \cos\left(\frac{1}{2}x^2\right)$$

$$\begin{aligned} f(x) &= x \cos\left(\frac{1}{2}x^2\right) = x \sum_{n=0}^{\infty} (-1)^n \frac{\left(\frac{1}{2}x^2\right)^{2n}}{(2n)!} \\ &= x \sum_{n=0}^{\infty} (-1)^n \frac{\frac{1}{2^{2n}} x^{4n}}{(2n)!} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{4n+1}}{2^{2n} (2n)!} \end{aligned}$$

The radius of convergence :
 $R = \infty$

Find the Maclaurin series of the given function

$$f(x) = \sin^2 x$$

$$\begin{aligned} f(x) &= \frac{1}{2} (1 - \cos(2x)) \\ &= \frac{1}{2} \left(1 - \sum_{n=0}^{\infty} (-1)^n \frac{(2x)^{2n}}{(2n)!} \right) \\ &= \frac{1}{2} \left(1 - \left[1 + \sum_{n=1}^{\infty} (-1)^n \frac{2^{2n} x^{2n}}{(2n)!} \right] \right) \quad \text{where } n=0 \rightarrow a_0=1 \\ &= \frac{1}{2} \sum_{n=1}^{\infty} (-1)^n \frac{2^{2n} x^{2n}}{(2n)!} = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2^{2n-1} x^{2n}}{(2n)!} \end{aligned}$$

$R = \infty$

Find the power series representation for

$$f(x) = \sin x \cos x$$

$$f(x) = \sin x \cos x$$

$$\longrightarrow \sin(2x) = 2 \sin x \cos x$$

$$\frac{1}{2} \sin(2x) = \sin x \cos x$$

$$f(x) = \frac{1}{2} \sin(2x)$$

$$= \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \frac{(2x)^{2n+1}}{(2n+1)!}$$

$$\Rightarrow R = \infty$$

Find the power series representation for

$$f(x) = x \sin^2(4x)$$

$$f(x) = x \cdot \frac{1}{2} [1 - \cos(8x)]$$

$$= \frac{x}{2} \left[1 - \sum_{n=0}^{\infty} (-1)^n \frac{(8x)^{2n}}{(2n)!} \right]$$

$$\xrightarrow{n=0} (-1)^0 \frac{(8x)^0}{0!} = 1$$

$$= \frac{x}{2} \left[1 - \left(1 + \sum_{n=1}^{\infty} (-1)^n \frac{8^{2n} x^{2n}}{(2n)!} \right) \right]$$

$$= \frac{x}{2} \left[1 - 1 - \sum_{n=1}^{\infty} (-1)^n \frac{(2^3)^{2n} x^{2n}}{(2n)!} \right]$$

$$= -\frac{x}{2} \sum_{n=1}^{\infty} (-1)^n \frac{2^{6n} x^{2n}}{(2n)!}$$

$$= \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2^{6n-1} x^{2n+1}}{(2n)!}$$

$$\Rightarrow R = \infty$$

Find the Maclaurin series for $f(x) = 5^x$

and the radius of convergence.

May 2017

$$f(x) = 5^x$$

$$= e^{\ln 5^x} = e^{x \ln 5} = \sum_{n=0}^{\infty} \frac{(x \ln 5)^n}{n!}$$

radius of convergence : $R = \infty$

Find the power series representation for

$$f(x) = 2^x \text{ about } x = 1$$

and determine the radius of convergence

$$x - 1 = 0$$

$$\text{Let } t = x - 1 \rightarrow x = t + 1$$

$$f(t) = 2^{t+1} = 2^t \cdot 2^1 = 2 e^{\ln 2^t}$$

$$= 2 e^{t \ln 2}$$

$$= 2 \sum_{n=0}^{\infty} \frac{(t \ln 2)^n}{n!}$$

$$= 2 \sum_{n=0}^{\infty} \frac{((x-1) \ln 2)^n}{n!}$$

$$R = \infty$$

Let f be the continuous function satisfying

$$f(0) = 4, \text{ and } f'(x) = \frac{\sin x}{x} \text{ for } x \neq 0.$$

Find a power series representation of f .

Past exam

$$f'(x) = \frac{1}{x} \sin x = u$$

$$= \frac{1}{x} \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n+1)!}$$

$$\begin{aligned} \therefore f(x) &= \int \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n+1)!} dx \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)(2n+1)!} + C \end{aligned}$$

at $x=0$

$$f(0) = 0 + C \rightarrow 4 = C$$

$$\therefore f(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)(2n+1)!} + 4$$

Let $f(x) = \frac{e^x - 1}{x}$

Find the power series representation of the derivative
then use it to find the sum of

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!}$$

$$f(x) = \frac{1}{x}(e^x - 1) = \frac{1}{x} \left[\sum_{n=0}^{\infty} \frac{x^n}{n!} - 1 \right]$$

أبواب بالذات للتفاضل من الجمع والطرح.

$$= \frac{1}{x} \left[1 + \sum_{n=1}^{\infty} \frac{x^n}{n!} - 1 \right]$$

$$= \frac{1}{x} \sum_{n=1}^{\infty} \frac{x^n}{n!}$$

$$\frac{1}{x}(e^x - 1) = \sum_{n=1}^{\infty} \frac{x^{n-1}}{n!}$$

Diff. both sides w.r.t. x

$$\frac{1}{x^2}(e^x - 1) + \frac{1}{x}e^x = \sum_{n=2}^{\infty} \frac{(n-1)x^{n-2}}{n!}$$

بقولها

$$n \rightarrow n+1$$

$$= \sum_{n+1=2}^{\infty} \frac{(n+1-1)x^{n+1-2}}{(n+1)!}$$

$$= \sum_{n=1}^{\infty} \frac{n x^{n-1}}{(n+1)!}$$

at $x=1$

$$-1(e^1 - 1) + 1e^1 = \sum_{n=1}^{\infty} \frac{n}{(n+1)!}$$

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!}$$

$$x^{n-1} = 1$$

$x=1$

Let $f(x) = x^3 e^{\frac{x}{2}}$

Find the power series representation of $f(x)$.

then use it to find the sum of

$$\sum_{n=0}^{\infty} \frac{n+3}{2^n n!}$$

$$\begin{aligned} f(x) &= x^3 e^{\frac{x}{2}} = x^3 \sum_{n=0}^{\infty} \frac{(\frac{x}{2})^n}{n!} \\ &= x^3 \sum_{n=0}^{\infty} \frac{\frac{1}{2^n} x^n}{n!} \\ &= \sum_{n=0}^{\infty} \frac{x^{n+3}}{2^n n!} \end{aligned}$$

* ما قبل السطر

Diff. both sides w.r.t. x

$$3x^2 e^{\frac{x}{2}} + x^3 e^{\frac{x}{2}} \cdot \frac{1}{2} = \sum_{n=0}^{\infty} \frac{(n+3) x^{n+2}}{2^n n!}$$

at $x=1$

$$3e^{\frac{1}{2}} + e^{\frac{1}{2}} \cdot \frac{1}{2} = \sum_{n=0}^{\infty} \frac{(n+3)}{2^n n!}$$

$$\frac{6e^{\frac{1}{2}}}{2} + \frac{e^{\frac{1}{2}}}{2}$$

$$\frac{7e^{\frac{1}{2}}}{2}$$

$$= \sum_{n=0}^{\infty} \frac{(n+3)}{2^n n!}$$

$$\sum_{n=0}^{\infty} \frac{n+3}{2^n n!}$$

$$x^{n+2} = 1$$

$$x=1$$

a) Find the Maclaurin series for e^{-x} and its radius of convergence.

b) Find the sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}n!}$

Dec 2017

$$(a) e^{-x} = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!}$$

radius of conv
 $R = \infty$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}n!}$$

$$x^n = \frac{1}{2^{n+1}}$$

$$x^n = \frac{1}{2^n \cdot 2}$$

!!

Now

$$\frac{1}{2} x^n = \frac{1}{2^{n+1}}$$

$$\frac{1}{2} x^n = \frac{1}{2^n \cdot 2}$$

$$x^n = \frac{1}{2^n}$$

$$x^n = \left(\frac{1}{2}\right)^n$$

$$\times \frac{1}{2}$$

$$\frac{1}{2} e^{-x} = \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n \left(\frac{1}{2}\right)^n}{n!}$$

$$\therefore \text{at } x = \frac{1}{2}$$

$$\frac{1}{2} e^{-\frac{1}{2}} = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}n!}$$

Find the sum of the series

$$\sum_{n=0}^{\infty} \frac{[-\ln 4]^n}{2^n n!}$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\sum_{n=0}^{\infty} \frac{[-\ln 4]^n}{2^n n!}$$

$$\text{at } x = -\frac{1}{2} \ln 4$$

$$x^n = \frac{[-\ln 4]^n}{2^n}$$

$$e^{-\frac{1}{2} \ln 4} = \sum_{n=0}^{\infty} \frac{[-\ln 4]^n}{2^n n!}$$

$$x^n = \left(-\frac{\ln 4}{2}\right)^n$$

$$e^{\ln 4 \cdot \frac{-1}{2}}$$

$$\therefore x = -\frac{1}{2} \ln 4$$

$$4^{-\frac{1}{2}} = \frac{1}{4^{\frac{1}{2}}} = \frac{1}{2}$$

$$\therefore \frac{1}{2} = \sum_{n=0}^{\infty} \frac{[-\ln 4]^n}{2^n n!}$$

Find the sum of the series

$$\sum_{n=0}^{\infty} \frac{[\ln 2]^{2n}}{(2n)!}$$

$$\cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

at $x = \ln 2$

$$\cosh \ln 2 = \sum_{n=0}^{\infty} \frac{[\ln 2]^{2n}}{(2n)!}$$

$$e^{\ln 2} + e^{-\ln 2}$$

$$\frac{e^{\ln 2} + e^{-\ln 2}}{2} = \frac{2 + \frac{1}{2}}{2} = \frac{\frac{5}{2}}{2} = \frac{5}{4}$$

$$\sum_{n=0}^{\infty} \frac{[\ln 2]^{2n}}{(2n)!}$$

$$x^{2n} = [\ln 2]^{2n}$$

$$x = \ln 2$$

Find the sum of the series

$$\sum_{n=0}^{\infty} \frac{[-\pi^2]^n}{9^n (2n)!}$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

at $x = \frac{\pi}{3}$

$$\cos \frac{\pi}{3} = \sum_{n=0}^{\infty} \frac{[-\pi^2]^n}{9^n (2n)!}$$

$$\frac{1}{2} = \sum_{n=0}^{\infty} \frac{[-\pi^2]^n}{9^n (2n)!}$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n}}{9^n (2n)!}$$

$$x^{2n} = \frac{\pi^{2n}}{9^n}$$

$$x^{2n} = \frac{\pi^{2n}}{3^{2n}}$$

$$x^{2n} = \left(\frac{\pi}{3}\right)^{2n}$$

$$\therefore x = \frac{\pi}{3}$$

Find the sum of the series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$$

$$\tan^{-1} x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

at $x=1$

$$\tan^{-1} 1 = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$$

$$\frac{\pi}{4} = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$$

$$x^{2n+1} = 1$$

$$x = 1$$

Find the sum of the series

$$1 - \ln 2 + \frac{(\ln 2)^2}{2!} - \frac{(\ln 2)^3}{3!} + \dots$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n (\ln 2)^n}{n!} \quad ??$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

at $x = -\ln 2$

$$e^{-\ln 2} = \sum_{n=0}^{\infty} \frac{(-1)^n (\ln 2)^n}{n!}$$

$$e^{\ln 2^{-1}} = 2^{-1} = \frac{1}{2}$$

$$\therefore \frac{1}{2} = \sum_{n=0}^{\infty} \frac{(-1)^n (\ln 2)^n}{n!}$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n (\ln 2)^n}{n!}$$

$$x^n = (-1)^n (\ln 2)^n$$

$$x^n = (-\ln 2)^n$$

$$x = -\ln 2$$

Find the Sum of the series for

$$\sum_{n=1}^{\infty} (-1)^n \frac{\pi^{2n+1}}{4^{2n+1} (2n+1)!}$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$n=0$$

$$(-1)^0 \frac{x^1}{(1)!} = x$$

$$\sin x = x + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$\sin x - x = \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$\text{at } x = \frac{\pi}{4}$$

$$\sin \frac{\pi}{4} - \frac{\pi}{4}$$

$$\frac{1}{\sqrt{2}} - \frac{\pi}{4}$$

$$= \sum_{n=1}^{\infty} (-1)^n \frac{\pi^{2n+1}}{4^{2n+1} (2n+1)!}$$

$$= \sum_{n=1}^{\infty} (-1)^n \frac{\pi^{2n+1}}{4^{2n+1} (2n+1)!}$$

$$\sum_{n=1}^{\infty} (-1)^n \frac{\pi^{2n+1}}{4^{2n+1} (2n+1)!}$$

$$x^{2n+1} = \frac{\pi^{2n+1}}{4^{2n+1}}$$

$$x^{2n+1} = \left(\frac{\pi}{4}\right)^{2n+1}$$

$$x = \frac{\pi}{4}$$



Definition



Maclurain series $\rightarrow x = 0$



$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots +$$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$



Taylor series $\rightarrow x = a$



$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots +$$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

Find the Taylor series for the function using definition

$$f(x) = (x-2)^2 e^{2-x} \text{ about } x=2$$

$$\text{Let } g(x) = e^{2-x} \longrightarrow g(2) = e^0 = 1$$

$$g'(x) = -e^{2-x} \longrightarrow g'(2) = -1$$

$$g''(x) = e^{2-x} \longrightarrow g''(2) = 1$$

$$g'''(x) = -e^{2-x} \longrightarrow g'''(2) = -1$$

$$\begin{aligned} \therefore g(x) &= g(2) + \frac{g'(2)}{1!} (x-2) + \frac{g''(2)}{2!} (x-2)^2 + \frac{g'''(2)}{3!} (x-2)^3 + \dots \\ &= 1 + \frac{-1}{1!} (x-2) + \frac{1}{2!} (x-2)^2 + \frac{-1}{3!} (x-2)^3 + \dots \end{aligned}$$

$$g(x) = \sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^n}{n!}$$

$$\begin{aligned} f(x) &= (x-2)^2 e^{2-x} \\ &= (x-2)^2 \sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^n}{n!} \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^{n+2}}{n!} \end{aligned}$$

Find the Maclaurin series for the function using definition

$$f(x) = \frac{x^2}{3^x}$$

$$f(x) = x^2 3^{-x}$$

$$\text{Let } g(x) = 3^{-x} \longrightarrow g(0) = 1$$

$$g'(x) = -3^{-x} \ln 3 \longrightarrow g'(0) = -\ln 3$$

$$g''(x) = 3^{-x} (\ln 3)^2 \longrightarrow g''(0) = (\ln 3)^2$$

$$g'''(x) = -3^{-x} (\ln 3)^3 \longrightarrow g'''(0) = -(\ln 3)^3$$

$$\therefore g(x) = g(0) + \frac{g'(0)}{1!} x + \frac{g''(0)}{2!} x^2 + \frac{g'''(0)}{3!} x^3 + \dots$$

$$= 1 + \frac{-\ln 3}{1!} x + \frac{(\ln 3)^2}{2!} x^2 + \frac{-(\ln 3)^3}{3!} x^3 + \dots$$

$$g(x) = \sum_{n=0}^{\infty} (-1)^n \frac{(\ln 3)^n}{n!} x^n$$

$$f(x) = x^2 g(x)$$

$$= x^2 \sum_{n=0}^{\infty} (-1)^n \frac{(\ln 3)^n}{n!} x^n$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{(\ln 3)^n}{n!} x^{n+2}$$

Find the Taylor series for the function using definition

$$f(x) = \cos x \text{ about } x = \frac{2\pi}{3}$$

$$f(x) = \cos x \longrightarrow f\left(\frac{2\pi}{3}\right) = \frac{1}{2}$$

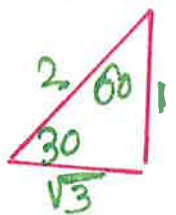
$$f'(x) = -\sin x \longrightarrow f'\left(\frac{2\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

$$f''(x) = -\cos x \longrightarrow f''\left(\frac{2\pi}{3}\right) = -\frac{1}{2}$$

$$f'''(x) = \sin x \longrightarrow f'''\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$f^{(4)}(x) = \cos x \longrightarrow f^{(4)}\left(\frac{2\pi}{3}\right) = \frac{1}{2}$$

$$f^{(5)}(x) = -\sin x \longrightarrow f^{(5)}\left(\frac{2\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$



$$\begin{aligned} f(x) = & f\left(\frac{2\pi}{3}\right) + \frac{f'\left(\frac{2\pi}{3}\right)}{1!} \left(x - \frac{2\pi}{3}\right) + \frac{f''\left(\frac{2\pi}{3}\right)}{2!} \left(x - \frac{2\pi}{3}\right)^2 \\ & + \frac{f'''\left(\frac{2\pi}{3}\right)}{3!} \left(x - \frac{2\pi}{3}\right)^3 + \frac{f^{(4)}\left(\frac{2\pi}{3}\right)}{4!} \left(x - \frac{2\pi}{3}\right)^4 \\ & + \frac{f^{(5)}\left(\frac{2\pi}{3}\right)}{5!} \left(x - \frac{2\pi}{3}\right)^5 + \dots \end{aligned}$$

See Next page plz!

$$\begin{aligned}
 &= -\frac{1}{2} + \frac{-\frac{\sqrt{3}}{2}}{1!} \left(x - \frac{2\pi}{3}\right) \\
 &+ \frac{\frac{1}{2}}{2!} \left(x - \frac{2\pi}{3}\right)^2 + \frac{\frac{\sqrt{3}}{2}}{3!} \left(x - \frac{2\pi}{3}\right)^3 \\
 &+ \frac{-\frac{1}{2}}{4!} \left(x - \frac{2\pi}{3}\right)^4 + \frac{-\frac{\sqrt{3}}{2}}{5!} \left(x - \frac{2\pi}{3}\right)^5 + \dots
 \end{aligned}$$

$$= \frac{1}{2} \sum_{n=0}^{\infty} (-1)^{n+1} \frac{\left(x - \frac{2\pi}{3}\right)^{2n}}{(2n)!} + \frac{\sqrt{3}}{2} \sum_{n=0}^{\infty} (-1)^{n+1} \frac{\left(x - \frac{2\pi}{3}\right)^{2n+1}}{(2n+1)!}$$

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Find the Taylor polynomial of degree 2 for

$$f(x) = \sqrt{1+x} \text{ about } x=0$$

$$f(x) = \sqrt{1+x} \longrightarrow f(0) = \sqrt{1} = 1$$

$$f'(x) = \frac{1}{2\sqrt{1+x}} \longrightarrow f'(0) = \frac{1}{2\sqrt{1+0}} = \frac{1}{2}$$

$$\begin{aligned} f''(x) &= \frac{0 - 2 \cdot \frac{1}{2\sqrt{1+x}}}{[2\sqrt{1+x}]^2} \\ &= \frac{-1}{4(1+x)\sqrt{1+x}} \longrightarrow f''(0) = \frac{-1}{4(1)\sqrt{1}} \\ &= -\frac{1}{4} \end{aligned}$$

$$\begin{aligned} T_2(x) &= f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 \\ &= 1 + \frac{\frac{1}{2}}{1}x + \frac{-\frac{1}{4}}{2(1)}x^2 \end{aligned}$$

Find the Taylor polynomial of degree two for

$$f(x) = \sin x \quad \text{at} \quad a = \frac{\pi}{4}$$

$$f(x) = \sin x \longrightarrow f\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$f'(x) = \cos x \longrightarrow f'\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$f''(x) = -\sin x \longrightarrow f''\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$

$$\begin{aligned} T_2(x) &= f\left(\frac{\pi}{4}\right) + \frac{f'\left(\frac{\pi}{4}\right)}{1!} \left(x - \frac{\pi}{4}\right) + \frac{f''\left(\frac{\pi}{4}\right)}{2!} \left(x - \frac{\pi}{4}\right)^2 \\ &= \frac{1}{\sqrt{2}} + \frac{\frac{1}{\sqrt{2}}}{1!} \left(x - \frac{\pi}{4}\right) + \frac{-\frac{1}{\sqrt{2}}}{2!} \left(x - \frac{\pi}{4}\right)^2 \end{aligned}$$

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Find the first 4 terms of the Taylor series of

$$f(x) = \sqrt{3+x^2} \text{ at } x = 1$$

Dec 2016

$$f(x) = \sqrt{3+x^2} \longrightarrow f(1) = \sqrt{4} = \boxed{2}$$

$$f'(x) = \frac{2x}{2\sqrt{3+x^2}} \longrightarrow f'(1) = \frac{1}{\sqrt{4}} = \boxed{\frac{1}{2}}$$

$$f''(x) = \frac{\sqrt{3+x^2}(1) - x \cdot \frac{2x}{2\sqrt{3+x^2}}}{3+x^2} = \frac{\frac{3+x^2-x^2}{\sqrt{3+x^2}}}{3+x^2} \longrightarrow f''(1) = \frac{3}{(4)^{\frac{3}{2}}} = \frac{3}{2^3} = \boxed{\frac{3}{8}}$$

$$f'''(x) = \frac{-3 \cdot \frac{3}{2} (3+x^2)^{-\frac{1}{2}} \cdot 2x}{(3+x^2)^3} \longrightarrow f'''(1) = \frac{-9(4)^{-\frac{1}{2}}(1)}{(4)^3} = \frac{-9(2)}{64} = \boxed{\frac{-9}{32}}$$

$$T_3(x) = f(1) + \frac{f'(1)}{1!}(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \frac{f'''(1)}{3!}(x-1)^3$$

$$= 2 + \frac{\frac{1}{2}}{1}(x-1) + \frac{\frac{3}{8}}{2!}(x-1)^2 + \frac{\frac{-9}{32}}{3!}(x-1)^3$$



Multiple choice



The third term of Maclaurin series of $(1+x)^{1/3}$ is

- a) equal to $(-2/9)x^2$ b) equal to $(-1/9)x^2$ c) equal to $(10/27)x^3$
d) equal to $(1/3)x^2$ e) equal to $(5/81)x^3$

Dec 2017

Let $f(x) = (1+x)^{1/3}$

$$f'(x) = \frac{1}{3} (1+x)^{-2/3}$$

$$f''(x) = \frac{1}{3} \cdot \frac{-2}{3} (1+x)^{-5/3}$$

$$= \frac{-2}{9} \frac{1}{(1+x)^{5/3}}$$

$$\begin{aligned} \longrightarrow f''(0) &= \frac{-2}{9} \frac{1}{(1+0)^{5/3}} \\ &= \frac{-2}{9} \end{aligned}$$

$$T_2(x) = f(0) + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2$$

$$\therefore \text{The third term is } \frac{\frac{-2}{9}}{2!} x^2$$

$$= \frac{-2}{2(9)} x^2$$

(b)



Multiple choice



The third term of the Taylor series for $f(x) = \frac{2}{x}$ centered at 2 is

a) $\ln 2$

b) $\frac{1}{8}(x-2)^3$

c) $\frac{1}{4}(x-2)^2$

d) $-\frac{1}{8}(x-2)^3$

e) $-\frac{1}{4}(x-2)^2$

May 2017

$$f(x) = \frac{2}{x}$$

$$f'(x) = \frac{-2}{x^2} = -2x^{-2}$$

$$f''(x) = 4x^{-3} = \frac{4}{x^3} \rightarrow f''(2) = \frac{4}{(2)^3} = \frac{4}{8} = \frac{1}{2}$$

$$T_2(x) = f(2) + \frac{f'(2)}{1!}(x-2) + \frac{f''(2)}{2!}(x-2)^2$$

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$$\therefore \text{The third term} = \frac{\frac{1}{2}}{2!}(x-2)^2$$

$$= \frac{1}{4}(x-2)^2$$

(c)

Let $f(x) = \frac{x}{2+x^3}$

a) Find the Maclaurin series for $f(x)$ and state the radius of convergence.

b) Use the Maclaurin series in (a) to find $f^{(31)}(0)$.

$$\frac{x}{2+x^3} = \frac{x}{2} \cdot \frac{1}{1+\frac{x^3}{2}} = \frac{x}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{x^3}{2}\right)^n$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{3n+1}}{2^{n+1}}$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

$$\left|\frac{x^3}{2}\right| < 1$$

$$|x^3| < 2$$

$$|x| < \sqrt[3]{2}$$

$$R = \sqrt[3]{2}$$

$$3n+1 = 31$$

$$3n = 30$$

$$n = 10$$

$$n = 31$$

$$\frac{(-1)^{10}}{2^{11}} = \frac{f^{(31)}(0)}{(31)!}$$

$$\frac{1}{2^{11}} = \frac{f^{(31)}(0)}{(31)!}$$

$$\therefore f^{(31)}(0) = \frac{(31)!}{2^{11}}$$

Find the Maclaurin series of $f(x) = \frac{x^2}{1+x^3}$

then find $f^{(29)}(0)$

Past exam

$$\begin{aligned} & \frac{x^2}{1+x^3} \\ & x^2 \frac{1}{1+x^3} \\ & x^2 \sum_{n=0}^{\infty} (-1)^n (x^3)^n \\ & \sum_{n=0}^{\infty} (-1)^n x^{3n+2} \\ & |x^3| < 1 \\ & |x| < 1 \\ & R=1 \\ & f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n \\ & 3n+2 = 29 \\ & 3n = 27 \\ & n = 9 \\ & \Rightarrow (-1)^9 x^{29} = \frac{f^{(29)}(0)}{(29)!} x^{29} \\ & -1 = \frac{f^{(29)}(0)}{(29)!} \\ & \therefore f^{(29)}(0) = -(29)! \end{aligned}$$

Let $f(x) = \frac{x}{(x-3)^2}$

(a) Find a power series representation of f centered at 0.

(b) Find $f^{(10)}(0)$.

Dec 2015

$$\frac{1}{x-3} = \frac{1}{-3} \frac{1}{1 - \frac{x}{3}} = -\frac{1}{3} \sum_{n=0}^{\infty} \left(\frac{x}{3}\right)^n$$

$$\frac{1}{x-3} = - \sum_{n=0}^{\infty} \frac{x^n}{3^{n+1}}$$

Diff. w.r.t. x

$$\frac{-1}{(x-3)^2} = - \sum_{n=1}^{\infty} \frac{n x^{n-1}}{3^{n+1}}$$

* (-x)

$$\frac{x}{(x-3)^2} = x \sum_{n=1}^{\infty} \frac{n x^{n-1}}{3^{n+1}}$$

$$\frac{x}{(x-3)^2} = \sum_{n=1}^{\infty} \frac{n x^n}{3^{n+1}}$$

$$\frac{x}{(x-3)^2} = \sum_{n=0}^{\infty} \frac{(n+1) x^{n+1}}{3^{n+2}}$$

$$\begin{aligned} n+1 &= 10 \\ n &= 9 \end{aligned}$$

$$\Rightarrow \frac{10}{3^{11}} = \frac{f^{(10)}(0)}{(10)!}$$

$$\therefore f^{(10)}(0) = \frac{10}{3^{11}} (10)!$$

$$\left| \frac{x}{3} \right| < 1$$

$$|x| < 3$$

$$R=3$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

$$n=10$$

$$\text{Let } f(x) = \frac{x^3}{2-x} = \sum_{n=0}^{\infty} \frac{x^{n+3}}{2^{n+1}}$$

a) Find $f^{(12)}(0)$.

b) Show that $\sum_{n=0}^{\infty} \frac{(n+3)}{2^{n+1}} = 4$

$$\frac{x^3}{2-x} = \sum_{n=0}^{\infty} \frac{x^{n+3}}{2^{n+1}}$$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

$$n+3=12$$

$$n=9$$

$$n=12$$

$\Rightarrow$

$$\frac{x^{n+3}}{2^{n+1}} = \frac{f^{(n)}(0)}{n!} x^n$$

$$\frac{x^{12}}{2^{10}} = \frac{f^{(12)}(0)}{(12)!} x^{12}$$

$$\frac{f^{(12)}(0)}{(12)!} = \frac{1}{2^{10}}$$

$$f^{(12)}(0) = \frac{(12)!}{2^{10}}$$

$$\frac{x^3}{(2-x)} = \sum_{n=0}^{\infty} \frac{x^{n+3}}{2^{n+1}}$$

$$\sum_{n=0}^{\infty} \frac{(n+3)}{2^{n+1}}$$

Diff. w.r.t. x

$$\frac{(2-x) 3x^2 - x^3 (-1)}{(2-x)^2} = \sum_{n=0}^{\infty} \frac{(n+3)x^{n+2}}{2^{n+1}}$$

$$x^{n+2} = 1$$

$$x = 1$$

at $x = 1$

$$\frac{(2-1) 3(1) - (-1)}{(2-1)^2} = \sum_{n=0}^{\infty} \frac{(n+3)}{2^{n+1}}$$

$$\frac{(1) 3 + 1}{1} = \sum_{n=0}^{\infty} \frac{(n+3)}{2^{n+1}}$$

$$4 = \sum_{n=0}^{\infty} \frac{(n+3)}{2^{n+1}}$$



Text book question



Use series to evaluate the limit.

$$\lim_{x \rightarrow 0} \frac{x - \ln(1+x)}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{x - \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{x - \left[\overset{n=0}{a_0=x} - \overset{n=1}{a_1=\frac{x^2}{2}} + \overset{n=2}{a_2=\frac{x^3}{3}} - \overset{n=3}{a_3=\frac{x^4}{4}} + \dots \right]}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{\cancel{x} - \cancel{x} + \frac{x^2}{2} - \frac{x^3}{3} + \frac{x^4}{4} - \dots}{x^2}$$

$$\lim_{x \rightarrow 0} \left[\frac{\frac{x^2}{2}}{x^2} - \frac{\frac{x^3}{3}}{x^2} + \frac{\frac{x^4}{4}}{x^2} - \dots \right]$$

$$\lim_{x \rightarrow 0} \left[\frac{1}{2} - \frac{x}{3} + \frac{x^2}{4} - \dots \right]$$

$$= \frac{1}{2} - 0 + 0 - \dots = \frac{1}{2}$$



Text book question



Use series to evaluate the limit.

$$\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{1}{6}x^3}{x^5}$$

$$\lim_{x \rightarrow 0} \frac{\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} - x + \frac{1}{6}x^3}{x^5}$$

$$\lim_{x \rightarrow 0} \frac{\cancel{x} - \frac{\cancel{x^3}}{3!} + \frac{x^5}{5!} - \frac{\cancel{x^7}}{7!} + \dots - \cancel{x} + \frac{1}{6}\cancel{x^3}}{x^5}$$

$3(2)(1) = 6$

$$\lim_{x \rightarrow 0} \left[\frac{\frac{x^5}{5!}}{x^5} - \frac{\frac{x^7}{7!}}{x^5} + \dots \right]$$

$$\lim_{x \rightarrow 0} \left[\frac{1}{5!} - \frac{x^2}{7!} + \dots \right]$$

$$\frac{1}{5!} - 0 + 0 + \dots$$

$$= \frac{1}{5!}$$



The Binomial Series



Form

$$(1 + u)^k = \sum_{n=0}^{\infty} \binom{k}{n} u^n$$

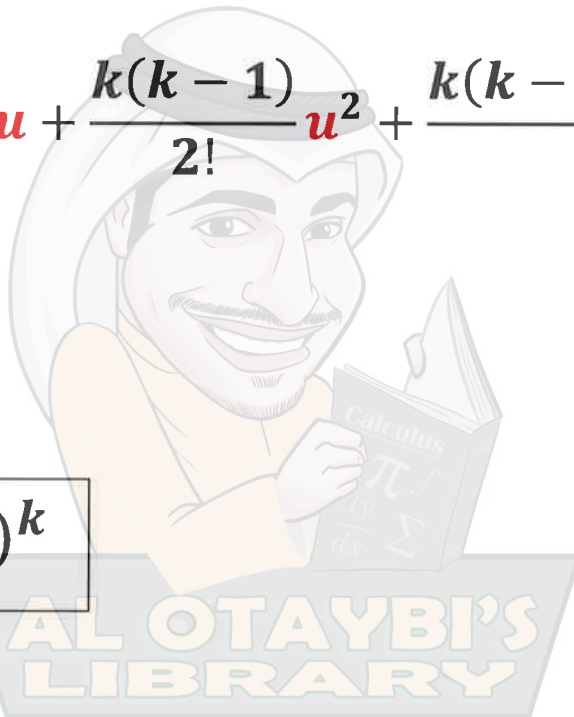
$$|u| < 1$$

$$= 1 + ku + \frac{k(k-1)}{2!} u^2 + \frac{k(k-1)(k-2)}{3!} u^3 + \dots$$



Sum

$$(1 + u)^k$$



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Coefficient of $\binom{k}{n}$

$$\frac{k(k-1)(k-2) \dots}{n!}$$



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Find the power series representation of

$$f(x) = \frac{1}{\sqrt{4-x}}$$

$$\begin{aligned} & [4-x]^{-\frac{1}{2}} = \left[4\left(1-\frac{x}{4}\right)\right]^{-\frac{1}{2}} = 4^{-\frac{1}{2}} \left(1-\frac{x}{4}\right)^{-\frac{1}{2}} \\ &= \frac{1}{4^{\frac{1}{2}}} \left(1-\frac{x}{4}\right)^{-\frac{1}{2}} \\ &= \frac{1}{2} \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} \left(\frac{-x}{4}\right)^n \\ &= \frac{1}{2} \left[1 + \binom{-\frac{1}{2}}{1} \left(\frac{-x}{4}\right) + \frac{\binom{-\frac{1}{2}}{2} \binom{-\frac{1}{2}-1}{2}}{2!} \left(\frac{-x}{4}\right)^2 + \dots \right] \end{aligned}$$

radius of convergence

$$\left| \frac{-x}{4} \right| < 1$$

$$\left| \frac{x}{4} \right| < 1$$

$$|x| < 4$$

$$R = 4$$

Find the sum of

$$\sum_{n=0}^{\infty} \binom{0.5}{n} \left(\frac{9}{16}\right)^n$$

$$\text{Sum} = (1+u)^k$$

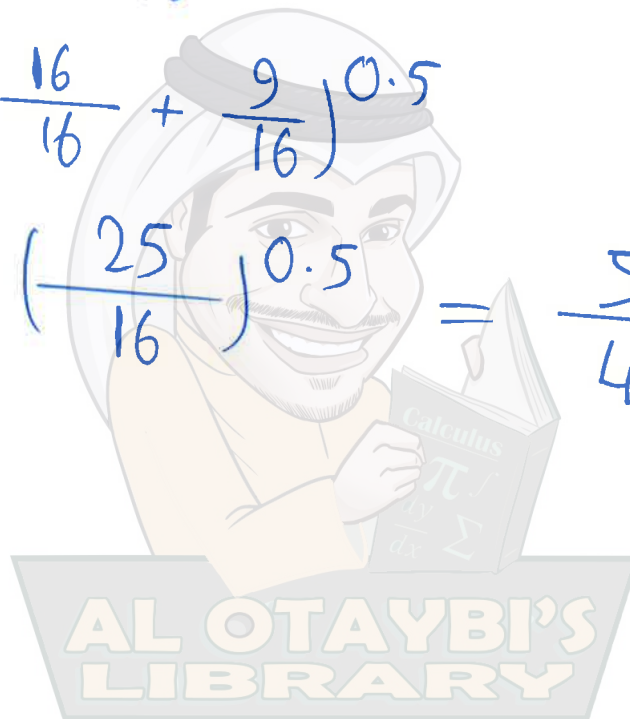
$$k = 0.5$$

$$u = \frac{9}{16}$$

$$= \left(1 + \frac{9}{16}\right)^{0.5}$$

$$= \left(\frac{16}{16} + \frac{9}{16}\right)^{0.5}$$

$$= \left(\frac{25}{16}\right)^{0.5} = \frac{5}{4}$$





Multiple choice



What is the sum of the series $\sum_{n=0}^{\infty} \binom{-1}{n} \frac{1}{2^n}$?

- a) $-\frac{2}{3}$ b) $\frac{2}{3}$ c) -2 d) 2 e) 1

May 2017

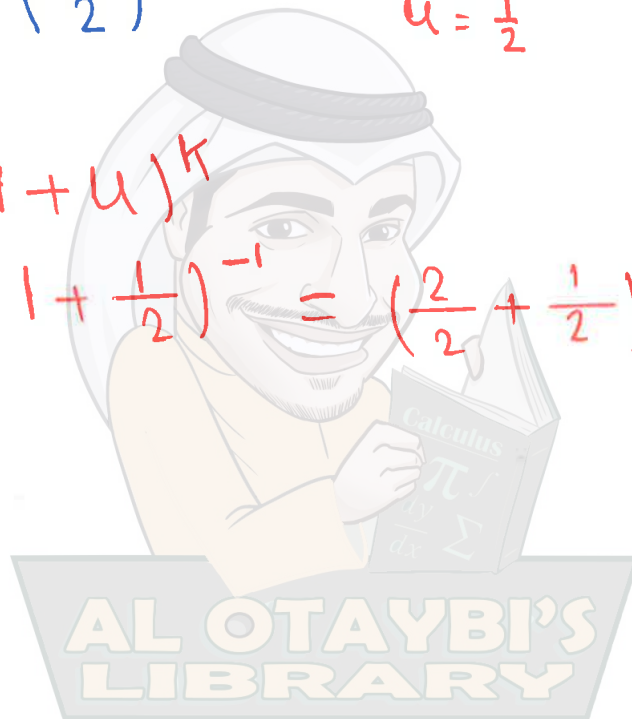
$$\sum_{n=0}^{\infty} \binom{-1}{n} \left(\frac{1}{2}\right)^n$$

$$k = -1 \\ u = \frac{1}{2}$$

(binomial series)

$$\begin{aligned} \text{Sum} &= (1+u)^k \\ &= \left(1 + \frac{1}{2}\right)^{-1} = \left(\frac{2}{2} + \frac{1}{2}\right)^{-1} = \left(\frac{3}{2}\right)^{-1} = \frac{2}{3} \end{aligned}$$

✓ (b)



Find the binomial coefficient $\binom{9}{4}$

$$\begin{aligned}
 \text{binomial coeff.} &= \frac{9(9-1)(9-2)(9-3)}{4!} \\
 &= \frac{9(8)(7)(6)}{4!} \\
 &= \frac{9(8)(7)(6)}{4(3)(2)(1)} = 126
 \end{aligned}$$



Multiple choice



The binomial coefficient $\binom{5/2}{3}$ equals

Dec 2016

a) $125/8$ b) $5/6$ c) $-5/9$ d) $5/16$ e) $\sqrt{3^5}$

$$\begin{aligned}
 \text{binomial coeff.} &= \frac{\left(\frac{5}{2}\right)\left(\frac{5}{2}-1\right)\left(\frac{5}{2}-2\right)}{3!} \\
 &= \frac{\frac{5}{2}\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)}{3!} \\
 &= \frac{\frac{15}{8}}{3(2)} = \frac{15}{8(3)(2)} = \frac{5}{16}
 \end{aligned}$$



Multiple choice



The binomial coefficient $\binom{-1}{4}$ equals to

- a) -1 b) $4!$ c) 1 d) $-\frac{1}{4!}$ e) $-5!$

Dec 2017

$$\text{binomial coeff.} = \frac{-1 \cdot (-1-1) \cdot (-1-2) \cdot (-1-3)}{4!}$$

$$= \frac{-1 \cdot (-2) \cdot (-3) \cdot (-4)}{4 \cdot (3) \cdot (2) \cdot (1)} = 1$$

(c)

